Installing Mbed OS Nano boards

A step-by-step guide to install the core needed for the Nano 33 BLE, Nano 33 BLE Sense and Nano RP2040 Connect boards.

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Installing Mbed OS Nano boards

In this tutorial, we will go through a few simple steps on installing the core needed for **Mbed OS Nano** boards. This installation is necessary to use your board with the offline Arduino IDE.

This tutorial uses the **Arduino IDE**. You can download the editor easily from our software page.

If you are using the **Cloud Editor**, you can follow the getting started with the Cloud Editor tutorial.

Boards using the Mbed core

- Arduino Nano 33 BLE
- Arduino Nano 33 BLE Sense
- Arduino Nano 33 BLE Sense Rev2
- Arduino Nano RP2040

Help

Downloading and installing

1. First, we need to download the Arduino IDE, which can be done from the software page.
2. Install the Arduino IDE on your local machine.
3. Open the Arduino IDE.

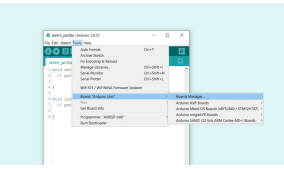
Installing the Mbed OS core for Nano boards

After we have downloaded, installed and opened the Arduino IDE, let's continue to install the core needed.

When we open the editor, we will see an empty sketch.

An empty Arduino IDE sketch window.

Here we need to navigate to **Tools > Board > Board Manager**.



Selecting board manager.

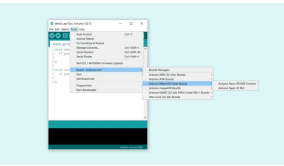
This will open up a new window, with all available cores. Find the one named **Arduino Mbed OS Nano Boards** and install it.

List of cores.

This process may take some time, and you may need to accept the installation window that comes up (depending on your operative system). When it is finished, it should say `"INSTALLED"` under the title.

> *Note: This process may take several minutes.*

Exit the board manager, and go to **Tools > Board > Arduino > Arduino Mbed OS Nano Boards**. Here you can see all the Mbed boards listed, where you can select the board you are using. You have now successfully installed the core.

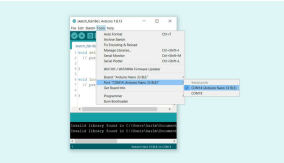


List of available boards.

Selecting the port

Now, let's make sure that our board is found by our computer, by selecting the port. Regardless

always need to choose the port for the board we are using. This is simply done by navigating to **Tools > Port**, where you select your board from the list.



Selecting the right board and port.

This will look different depending on what kind of operative system you are using.

For **Windows** users, it could look like this:

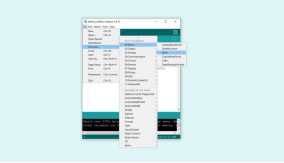
```
<COM14> (Arduino Nano BLE)
```

For **MAC** users, it could look like this:

```
/dev/cu.usbmodem14112 (Arduino Nano BLE)
```

Uploading a simple example

You are now ready to start using your board! The easiest way to check that everything is working, is to upload just a simple blink example to your board. This is done by navigating to **File > Examples > 01.Basics > Blink**.



Selecting the blink

To upload the sketch, simply click on the arrow in the top left corner. This process takes a few seconds, and it is important to not disconnect the board during this process.

Uploading the sketch.

When the code is uploaded, the text `"Done uploading."` is visible in the bottom left corner.

If you look closely at your board, you will notice an orange LED blink with an interval of one second. This means you have successfully uploaded a program to your board.

Suggest changes

The content on docs.arduino.cc is facilitated through a public GitHub repository. If you see anything wrong, you can edit this page here.

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